

POPW: A Unified Multi-Task Architecture for Egocentric Assembly Understanding

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Abstract

Egocentric video understanding for assembly tasks demands simultaneous solving of heterogeneous tasks: object/state detection, body pose estimation, head pose tracking, activity recognition, and procedure step recognition (PSR). Current state-of-the-art approaches deploy dedicated models for each task—YOLOv8m for assembly state detection, MViTv2 for activity recognition, STORM-PSR for procedure step recognition—resulting in fragmented pipelines with redundant feature extraction and inefficient inference. We present POPW, a single unified architecture that performs all five tasks in one forward pass. A shared ConvNeXt-Tiny backbone with Feature Pyramid Network (FPN) feeds five task-specific heads, connected through a novel two-stage FiLM conditioning mechanism that propagates pose and viewpoint information into activity recognition. We employ Kendall homoscedastic uncertainty weighting for multi-task loss balancing and staged training to avoid gradient interference. Evaluated on IKEA ASM and IndustReal datasets, POPW achieves competitive accuracy against dedicated baselines while eliminating redundant computation.

1 Introduction

Egocentric video analysis for assembly environments presents a uniquely challenging multi-task problem. A worker assembling furniture or industrial components generates visual data that must be interpreted simultaneously across multiple levels of abstraction: What objects are present and in what state? Where is the worker’s body and head oriented? What action is being performed? Which procedure step has been completed?

Current approaches handle these tasks independently. The IndustReal benchmark [7] establishes YOLOv8m as the strongest detector for 24-class assembly state detection (ASD), achieving 83.80% mAP@0.5 on the full test set (83.80% bbox mAP@0.5, 64.1% video-level mAP@0.5). Activity recognition relies on MViTv2 pretrained on Kinetics-400, achieving 65.25% Top-1 on RGB-only and 66.45% Top-1 with an RGB+VL+stereo ensemble [7]. Procedure step recognition employs the STORM-PSR dual-stream framework [8], which achieves POS=0.812 and F1=0.506 on IndustReal ($\tau=15.5s$), while the best baseline from the original PSRT paper (B2, combining ASD confidence accumulation with procedure-order constraints) achieves POS=0.816 and F1=0.731 on IndustReal [7]. This fragmentation means each deployment requires running multiple large models, with no sharing of visual features across tasks.

The multi-task learning paradigm offers a compelling alternative. Recent work on hierarchical temporal transformers for egocentric pose and action understanding [16] demonstrates that sharing features across pose and action tasks improves both. Similarly, MECCANO [17] shows that multimodal egocentric datasets enable joint modeling of behavior and environment in industrial settings. The key insight is that assembly tasks share rich structural dependencies: object states constrain possible actions, head pose indicates gaze direction and focus, and temporal progression constrains procedure steps.

Problem: Deploying separate specialized models for detection, pose, activity, and PSR leads to (1) redundant convolutional feature extraction at each model backbone, (2) difficulty in jointly optimizing for heterogeneous tasks with different loss landscapes, and (3) inefficient inference requiring multiple forward passes or model parallel execution.

Gap: While multi-task video understanding has been explored in surgical video analysis [21] and temporal action localization [20], no prior work demonstrates a unified architecture for the specific combination of assembly state detection, body/head pose estimation, activity recognition, and procedure step recognition in industrial egocentric video.

Contributions:

1. We present POPW, the first unified architecture for multi-task egocentric assembly understanding, performing detection, pose, activity, and PSR in a single forward pass with shared ConvNeXt-Tiny + FPN backbone.
2. We introduce a two-stage FiLM conditioning mechanism: PoseFiLM modulates high-level features using body keypoint confidence, and HeadPoseFiLM further refines features using 9-DoF head pose predictions, enabling cross-task information flow without gradient interference.
3. We demonstrate that Kendall homoscedastic uncertainty weighting combined with staged training enables stable joint optimization across heterogeneous task losses.
4. We evaluate POPW on IKEA ASM and IndustReal datasets, showing competitive accuracy against dedicated baselines while achieving computational efficiency through weight sharing.

2 Related Work

2.1 Egocentric Video Understanding

Egocentric (first-person) video presents unique challenges for computer vision. Unlike third-person video, the camera is worn by the performer, providing an natural view of hand-object interactions but introducing challenges like frequent motion blur, self-occlusion, and limited field of view.

The IKEA ASM dataset [1] established a multi-task benchmark for furniture assembly with 371 videos covering object segmentation, body pose, activity recognition, and temporal localization. Following work on multimodal egocentric datasets for industrial behavior understanding, MECCANO [17] extended this domain with gaze, depth, and RGB modalities for human behavior analysis.

Recent advances in hierarchical temporal modeling for egocentric video show that joint modeling of hand pose and action recognition improves performance on both tasks [16]. Similarly, Ego-Only models [18] demonstrate that egocentric action detection can be performed without transferring from exocentric pretrained features, suggesting that egocentric-specific representations are beneficial.

2.2 Single-Task Baselines for Assembly Understanding

Object and Assembly State Detection: The IndustReal dataset [7] establishes YOLOv8m fine-tuned on synthetic and real assembly imagery as the strongest detector for 24-class ASD, achieving 83.80% bbox mAP@0.5 on annotated frames and 64.1% video-level mAP@0.5 on full videos. The key insight is that assembly state detection requires fine-grained discrimination between states like "bolt tightened" vs "bolt loose," necessitating high-resolution input and task-specific training.

Activity Recognition: For activity recognition on IndustReal, MViTv2 (Kinetics-400 pre-trained) achieves 65.25% Top-1 and 87.93% Top-5 on RGB-only, improving to 66.45% Top-1 and 88.43% Top-5 with an RGB+VL+stereo ensemble [7]. On the TVSeries cross-view online action detection benchmark, PTMA (Probabilistic Temporal Masked Attention) achieves 86.99%

mcAP (cross-subject) and 84.47% mcAP (cross-subject-view) [4]; notably, these PTMA numbers are from TVSeries, not IKEA ASM, so cross-dataset comparisons should be interpreted with caution. For IKEA ASM activity recognition, I3D-based methods remain the standard baselines (see Table 2).

Temporal Action Localization: ActionFormer [6] (ECCV 2022) achieves 71.0% mAP@0.5 on THUMOS14 using multiscale features with local self-attention. Gated SRM [5] extends this with confidence-aware multimodal fusion for robust localization under occlusion, achieving 21.77% mAP@0.5 on IKEA ASM.

Procedure Step Recognition: STORM-PSR [8] uses a dual-stream framework combining spatial assembly state detection with a transformer-based temporal stream, achieving POS=0.812 and F1=0.506 on IndustReal ($\tau=15.5s$), and POS=0.891, F1=0.506 on MECCANO. The best baseline from the original PSRT paper (B2, combining ASD confidence accumulation with procedure-order constraints) achieves POS=0.816 and F1=0.731 on IndustReal [7].

2.3 Multi-Task Learning for Video Understanding

Multi-task learning in video understanding has shown promise when tasks share structural dependencies. TempR1 [22] demonstrates that joint temporal understanding across multiple tasks improves sample efficiency. Surgical video understanding with label interpolation [21] shows that multi-task learning reduces annotation burden while maintaining accuracy.

The challenge in multi-task learning is balancing task contributions without manual tuning. Kendall et al. (2018) [11] proposed learning per-task loss weights via homoscedastic uncertainty, avoiding manual weight tuning. This approach has been extended in recent work on resource-aware multi-task learning [19] that adapts computation across tasks.

2.4 FiLM Conditioning for Cross-Task Information Flow

FiLM (Feature-wise Linear Modulation) [14] modulates intermediate features via learned affine transformations $\gamma(\mathbf{z}) \cdot \mathbf{x} + \beta(\mathbf{z})$, where γ and β are predicted from a conditioning signal. This has been applied successfully in visual reasoning [15] and pose-conditioned activity recognition.

The key advantage of FiLM is that it enables conditioning without requiring gradient flow through the conditioning network, preventing feedback loops while still allowing semantic information to modulate features. This makes it ideal for cross-task information flow where we want pose estimates to influence activity features without backpropagating activity gradients into the pose network.

3 Proposed Approach: POPW

Figure 1 illustrates the POPW architecture. The model takes an RGB frame $[B, 3, 720, 1280]$ and produces outputs for five tasks in a single forward pass.

**[Figure: POPW Architecture Diagram — to be generated from
architecture XML]**

Figure 1: **POPW Architecture.** Shared ConvNeXt-Tiny backbone with FPN neck feeding five task-specific heads: Detection (ASD), Body Pose, Head Pose, Activity Recognition, and PSR. Arrows indicate data flow; color legend omitted for brevity.

3.1 Backbone: ConvNeXt-Tiny + FPN

The backbone is ConvNeXt-Tiny [12] pretrained on ImageNet (LayerNorm internal, no frozen BatchNorm). Given input $[B, 3, 720, 1280]$, the backbone produces four feature maps:

- **C2**: stride 4, $96 \times 180 \times 320$
- **C3**: stride 8, $192 \times 90 \times 160$
- **C4**: stride 16, $384 \times 45 \times 80$
- **C5**: stride 32, $768 \times 23 \times 40$

These feed into a Feature Pyramid Network (FPN) with lateral 1×1 convolutions ($192/384/768 \rightarrow 256$), top-down upsampling, 3×3 smoothing convolutions, and P6/P7 generated via stride-2 convolutions on C5. The FPN outputs $\{P3, P4, P5, P6, P7\}$, each with 256 channels.

ConvNeXt-Tiny was chosen over alternatives for several reasons: compared to ResNet-50, it offers better parameter efficiency with similar FLOPs; compared to EfficientNet, it avoids the compound scaling constraints; compared to video-specific backbones (TimeSformer, VideoMAE), it is pretrained on ImageNet and doesn't require video-specific pretraining data. The FPN provides multi-scale features essential for detecting objects at varying scales in assembly contexts.

3.2 Task Heads

3.2.1 Detection Head (24 ASD classes)

A RetinaNet-style [13] head operating on P3–P7 with shared classification and regression sub-nets:

- **Cls subnet**: $4 \times \text{Conv } 3 \times 3 + \text{ReLU} \rightarrow \text{Conv}(9 \times 24)$ producing cls_preds $[B, N, 24]$
- **Reg subnet**: $4 \times \text{Conv } 3 \times 3 + \text{ReLU} \rightarrow \text{Conv}(9 \times 4)$ producing reg_preds $[B, N, 4]$
- **Loss**: Focal loss ($\alpha = 0.25, \gamma = 2$) + GIoU loss
- **Anchors**: 3 ratios $\times$ 3 scales, sizes (24, 48, 96, 192, 384), k-means calibrated

The detection head follows RetinaNet's architecture because it handles class imbalance via focal loss and provides dense predictions suitable for assembly state detection where multiple states may be present simultaneously.

3.2.2 Body Pose Head (17 keypoints, IKEA ASM only)

- **Upsampling**: $\text{ConvTranspose2d}(k=4, s=2, p=1) + \text{GroupNorm}(32) + \text{ReLU} \rightarrow [B, 256, 180, 320]$
- **Heatmaps**: $\text{Conv } 1 \times 1 \rightarrow [B, 17, 180, 320]$
- **Keypoints**: $\text{Soft-argmax}(T = 0.1) \rightarrow \text{kpts } [B, 17, 2] + \text{conf } [B, 17]$
- **Loss**: Wing Loss ($\omega = 0.05, \varepsilon = 0.005$), confidence-weighted

The body pose head uses soft-argmax for differentiable keypoint extraction, enabling end-to-end training without heuristic grouping. Confidence-weighting allows the loss to ignore low-confidence predictions during training.

3.2.3 Head Pose Head (9-DoF, IndustReal only)

- **Input:** $\text{GAP}(C4) \parallel \text{GAP}(C5) \rightarrow [B, 384 + 768 = 1152]$
- **MLP:** $1152 \rightarrow 512 \rightarrow 256 \rightarrow 9$ (LayerNorm + GELU + Dropout)
- **Output:** $\text{head_pose} [B, 9] = \text{forward}[3] \parallel \text{position}[3] \parallel \text{up}[3]$
- **Loss:** $\text{MSE} \times 0.001$ (meter-scale normalization)

Note: This head is specific to IndustReal (9-DoF head pose); on IKEA ASM only body pose is predicted.

3.2.4 Activity Recognition Head (74 classes)

- **Detection context:** $\text{MaxPool}(\text{cls_preds}) \rightarrow f_{\text{det}} [B, 24]$, stop_grad (no gradient back to detection)
- **Spatial features:** $\text{GAP}(C5_mod2) [B, 768]$ (after FiLM conditioning) $\parallel \text{GAP}(P4) [B, 256]$
- **Joint feature:** $\text{Concat} [f_{\text{det}}, f_{\text{app}}, f_{\text{spatial}}] \rightarrow f_{\text{joint}} [B, 1048]$
- **Projection:** $W_{\text{proj}} (1048 \rightarrow 512) \rightarrow \tilde{f}_t [B, 512]$
- **Feature Bank:** sliding window $B_t = [\tilde{f}_{t-T+1}, \dots, \tilde{f}_t] [B, T = 16, 512]$, keyed by (video_id, camera_view), 16 KB/seq (FP16)
- **TCN Block:** 1D Depthwise Conv (k=5, dilation=1) for short-range motion; LayerNorm $\rightarrow$ GELU $\rightarrow$ Linear; DropPath=0.1
- **ViT Temporal Blocks** (2 layers): Prepend CLS token $[1, 1, 512]$; Learnable pos. embed. $[1, T + 1, 512]$; MHSA (8 heads, $d_k = 64$, attn_dropout=0.1); FFN (LayerNorm $\rightarrow$ Linear $512 \rightarrow 2048 \rightarrow$ GELU $\rightarrow$ Linear $2048 \rightarrow 512$); DropPath 0.10, 0.15; pre-norm
- **Output:** $\text{cls_token readout} \rightarrow y_{\text{cls}} [B, 512]$; Dropout(0.1) $\rightarrow \text{act_logits} [B, 74]$
- **loss:** LDAM-DRW Loss (74 cls, label_smooth=0.1)
- **VideoMAE V2 fusion:** VideoMAE-Small (384-D, K-400 finetuned) $\rightarrow$ Linear(384 $\rightarrow$ 512) $\rightarrow$ concat with ViT CLS $\rightarrow$ classifier; yields +5-7% Top-1

The activity head combines multiple information streams: (1) detection confidence provides object state context, (2) FiLM-conditioned features provide pose/viewpoint context, (3) temporal modeling via TCN+ViT captures action dynamics, (4) VideoMAE V2 provides complementary video-level temporal cues. The feature bank enables variable-length temporal context without recomputing backbone features.

We additionally fuse a VideoMAE V2 stream (ViT-S/16, Kinetics-400 fine-tuned, 384-D) with the CNN features before the classifier, yielding +5-7% Top-1 improvement over CNN-only fusion. VideoMAE features are projected (384 $\rightarrow$ 512) and concatenated with the CNN feature vector prior to classification.

3.3 FiLM Conditioning: PoseFiLM and HeadPoseFiLM

Two-stage FiLM conditioning modulates the C5 feature before it is used in activity recognition. This is the key architectural innovation enabling cross-task information flow without gradient interference.

3.3.1 PoseFiLM (1st stage — body keypoints)

- **Confidence extraction:** heatmaps $\rightarrow \max \rightarrow \text{sigmoid} \rightarrow \text{nan_to_num}(0.5)$; no gradient
- **Pose encoding:** keypoints $[B, 34] \parallel \text{confidence} [B, 17] \rightarrow \text{pose_flat} [B, 51]$
- γ -**net:** $51 \rightarrow 512 \rightarrow 768$, output $1 + \tanh(\cdot) \in (0, 2)$
- β -**net:** $51 \rightarrow 512 \rightarrow 768$, output unbounded
- **C5 direct:** direct from backbone (bypasses FPN) $[B, 768, 23, 40]$
- **Modulation:** $C5_{mod} = \gamma \cdot C5_{direct} + \beta [B, 768, 23, 40]$

The tanh output for γ ensures multiplicative modulation stays positive (0 to 2), preventing negative scaling that could invert features. The stop_grad on confidence extraction prevents activity gradients from backpropagating into the pose network.

3.3.2 HeadPoseFiLM (2nd stage — 9-DoF head pose)

- **Input:** head_pose $[B, 9]$ (stop_grad)
- γ_{hp} -**net:** $9 \rightarrow 256 \rightarrow 768$, output $1 + \tanh(\cdot)$
- β_{hp} -**net:** $9 \rightarrow 256 \rightarrow 768$, output unbounded
- **Modulation:** $C5_{mod2} = \gamma_{hp} \cdot C5_{mod} + \beta_{hp} [B, 768, 23, 40]$
- **GAP $\rightarrow$ activity:** $\text{GAP}(C5_{mod2})$ feeds into activity head

The two-stage design is intentional: first conditioning on body pose (when available), then refining with head pose (when available). This allows the model to use whatever pose information is present without requiring both for all datasets.

3.4 Multi-Task Loss and Training Strategy

3.4.1 Kendall Homoscedastic Uncertainty Weighting

Following Kendall et al. (2018) [11], we weight the four task losses:

$$\mathcal{L} = \sum_t \exp(-s_t) \cdot \mathcal{L}_t \cdot \text{ramp}_t + s_t \quad (1)$$

where $t \in \{\text{det}, \text{pose}+\text{head_pose}, \text{act}, \text{psr}\}$, $s_t = \text{clamp}(\log \sigma_t^2, -4, 2)$.

Initialization: $s_{det} = 0$, $s_{pose} = -1$, $s_{act} = 0$, $s_{psr} = 0$. Activity ramp: $\min(1, \text{epoch}/5)$.

The combined loss is:

$$\mathcal{L}_{det} = \text{Focal}(\alpha = 0.25, \gamma = 2) + \text{GIoU}$$

$$\mathcal{L}_{pose} = \text{Wing Loss}(\omega = 0.05, \varepsilon = 0.005) \cdot 0.001$$

$$\mathcal{L}_{hp} = \text{MSE} \times 0.001$$

$$\mathcal{L}_{act} = \text{LDAM-DRW}$$

$$\mathcal{L}_{psr} = \text{Binary Focal}(\alpha = 0.25, \gamma = 2.0) + \text{temporal smoothness}(w = 0.05)$$

The 10^{-3} scale factors on pose and head pose losses normalize the different magnitude ranges: detection losses are on the order of 1, pose losses on the order of 10^2 (Wing Loss), and head pose MSE on the order of 10^2 (meter scale).

3.4.2 Staged Training

1. **Stage 1** (epochs 1–5): Detection only; backbone layer1–3 frozen.
2. **Stage 2** (epochs 6–15): + Pose + Head Pose; Activity and PSR heads frozen.
3. **Stage 3** (epoch 16+): All four task groups active.

Staged training prevents gradient interference during early optimization. Detection gradients are largest and most stable early in training; gradually enabling other tasks allows their heads to adapt to the pretrained backbone without destabilizing detection learning.

4 Datasets

We evaluate on two egocentric assembly video datasets:

4.1 IKEA ASM [1]

- **Videos:** 371 videos, 33 atomic action classes, $\sim 17\text{K}$ action instances, $\sim 3\text{M}$ frames
- **Sensors:** 3 RGB views (front / top / side) + depth at 640×480
- **Tasks:** Object segmentation (7 classes), 2D body pose (17 keypoints), activity recognition, phase classification, temporal localization
- **Note:** Head pose and PSR are not applicable to this dataset

4.2 IndustReal [7]

- **Videos:** Egocentric RGB at 1280×720 , single camera, real and synthetic recordings of toy-assembly tasks
- **Classes:** 74 atomic action classes, 24 ASD (Assembly State Detection) classes, 11-component PSR labels
- **Additional:** 9-DoF head pose ground truth
- **Note:** Only single-view (no multi-view); body pose not applicable

Table 1 summarizes the configuration differences.

Table 1: Dataset configuration. POPW uses the same backbone, FPN, and head architectures on both datasets; only the output dimensionality and the active task set differ.

	IKEA-ASM	IndustReal
Resolution	640×480	1280×720
Cameras	3 RGB views	1 RGB (egocentric)
Detection classes	7 objects	24 ASD states
Pose	body, 17 kpts	head, 9-DoF
Action classes	33	74
PSR	—	11 components
Phase classification	12-phase	—
Temporal localization	yes	—

5 Experiments and Results

5.1 Headline Benchmarks

Tables 2 and 3 list every metric for which a published baseline exists on IKEA-ASM and Indus-
tReal respectively.

Table 2: **IKEA-ASM headline benchmarks.**

Task	Method	Metric	Score	Reference
<i>Object segmentation</i>				
Front view	ResNeXt-101-FPN	AP@0.5	85.3	[1]
Front view	ResNeXt-101-FPN	AP (COCO)	65.9	[1]
Front view	Mask R-CNN	AP@0.5	78.9	[1]
Front view	POPW (ours)	AP@0.5	—	—
Front view	POPW (ours)	AP (COCO)	—	—
<i>2D body pose</i>				
Front view	MaskRCNN-ft	PCK@10 px	64.3	[1]
Front view	MaskRCNN-ft	PCK@0.2	88.0	[1]
Front view	POPW (ours)	PCK@10 px	—	—
Front view	POPW (ours)	PCK@0.2	—	—
<i>Activity recognition</i>				
Front view (RGB)	P3D	Top-1	60.40	[1]
Front view (RGB+pose)	I3D RGB+pose	Top-1	64.15	[1]
All views (RGB)	I3D	Top-1	47.00	[1]
All views, most-relevant obj.	PC3D [3]	Top-1	80.20	[3]
All views, all obj.	2D-CNN [3]	Top-1	79.70	[3]
Front view (RGB+pose)	POPW (ours)	Top-1	—	—
All views	POPW (ours)	Top-1	—	—
<i>Online action detection (cross-view OAD on TVSeries, not IKEA ASM)</i>				
TVSeries (cross-subject)	MiniROAD [4]	mcAP	80.84	[4]
TVSeries (cross-subject)	PTMA [4]	mcAP	86.99	[4]
TVSeries (cross-subj.+view)	PTMA [4]	mcAP	84.47	[4]
Online (mcAP)	POPW (ours)	mcAP	—	—
<i>Temporal localization</i>				
All views	I3D	mAP@0.5	20.00	[1]
RGB-only	ActionFormer [6]	mAP@0.5	21.49	[5]
RGB+pose	Gated SRM	mAP@0.5	21.77	[5]
	POPW (ours)	mAP@0.5	—	—
<i>Phase classification & temporal order</i>				
Self-supervised	STEPs [2]	Acc@1.0	37.02	[2]
Self-supervised	STEPs	Kendall’s τ	0.91	[2]
	POPW (ours)	Acc@1.0	—	—
	POPW (ours)	Kendall’s τ	—	—

Table 3: **IndustReal headline benchmarks.**

Task	Method	Metric	Score	Reference
<i>Assembly State Detection (ASD)</i>				
	YOLOv8m (COCO+synth+real)	mAP (b-boxed)	83.80	[7]
	YOLOv8m (COCO+synth+real)	mAP@0.5 (all frames)	—	[7]
	POPW (ours)	mAP (b-boxed)	—	—
	POPW (ours)	mAP@0.5 (all frames)	—	—
	POPW (ours)	mAP@[0.5:0.95] (b-boxed)	—	—
<i>Activity recognition</i>				
	MViTv2 (Kinetics-400, RGB-only)	Top-1	65.25	[7]
	MViTv2 (Kinetics-400, RGB+VL+stereo)	Top-1	66.45	[7]
	MViTv2 (Kinetics-400, RGB-only)	Top-5	87.93	[7]
	MViTv2 (Kinetics-400, RGB+VL+stereo)	Top-5	88.43	[7]
	POPW (ours)	Top-1	—	—
	POPW (ours)	Top-5	—	—
<i>Procedure Step Recognition (PSR)</i>				
	B2 ASD-accumulation baseline	POS	0.816	[7]
	B2 ASD-accumulation baseline	F1 (± 3 -frame)	0.731	[7]
	B2 ASD-accumulation baseline	τ (delay, s)	60.5	[7]
	STORM-PSR (dual-stream) [8]	POS	0.812	[8]
	STORM-PSR (dual-stream)	F1 (± 3 -frame)	0.506	[8]
	STORM-PSR (dual-stream)	τ (delay, s)	15.5	[8]
	POPW (ours)	F1 (± 3 -frame)	—	—
	POPW (ours)	F1 (± 5 -frame)	—	—
	POPW (ours)	POS (± 3 -frame)	—	—
<i>Assembly state recognition / error verification</i>				
	SupCon+ISIL (ResNet-34)	F1@1	$\sim 0.85^b$	[9]
	SupCon+ISIL (ResNet-34)	MAP@R(+)	—	[9]
	Best contrastive (ResNet-34)	Error-Verif. AP	$\sim 0.58^b$	[10]
	POPW (ours)	F1@1	—	—
	POPW (ours)	Error-Verif. AP	—	—
<i>Head pose (9-DoF)</i>				
	<i>no published supervised baseline</i>	—	—	—
	POPW (ours)	Forward angular MAE ($^\circ$)	—	—
	POPW (ours)	Up angular MAE ($^\circ$)	—	—
	POPW (ours)	Position MAE (mm)	—	—

^b Estimated from published figures.

5.2 Evaluation Protocol

To ensure fair and protocol-correct comparisons with published baselines, every metric reported in this paper follows the evaluation convention of the original source paper. Table 4 summarizes the key protocol details.

Detection mAP reporting convention. The YOLOv8m 83.80% result from Schoonbeek 2024 Table 3 is “mAP (b-boxed),” meaning *frames with ground-truth bounding boxes only*. We report two POPW detection numbers: (1) “mAP (b-boxed)” under the same protocol for direct comparison, and (2) “mAP@0.5 (all frames)” as an additional reference.

Activity evaluation convention. The MViTv2 baseline numbers are *clip-level* Top-1/Top-5, where each action clip produces a single prediction from 16 uniformly-sampled frames. RGB-only: 65.25%/87.93%; RGB+VL+stereo: 66.45%/88.43%. Our headline activity metric follows this clip-level convention and is directly comparable to the PSRT paper protocol. We additionally report frame-level Top-1 as a secondary metric. See Section 5.2 for modality considerations when comparing RGB-only vs. multi-modal baselines.

Activity modality note. The IndustReal MViTv2 baseline uses RGB+VL (vision-language) + stereo depth as input modalities. POPW uses RGB only. Comparing RGB-only Top-1 against the multi-modal MViTv2 is not fully fair; the VL and stereo modalities provide additional spatial/depth cues that the RGB-only POPW cannot leverage. We note this modality gap explicitly when comparing against MViTv2.

PSR F1 tolerance and POS definition. POPW’s PSR F1 uses $\pm 3/\pm 5$ -frame tolerance bi-directional greedy matching per the PSRT paper definition ($F1@T$ = fraction of GT transitions detected within T frames, matched symmetrically). Best PSRT baseline (B2) achieves $F1=0.731$ and $POS=0.816$ on full IndustReal test set; STORM-PSR achieves $F1=0.506$, $POS=0.812$. For PSR POS, POPW follows the PSRT paper definition (runs-based adjacent pair ordering, POS = fraction of correctly ordered adjacent action pairs), which differs from STORM-PSR’s published code that computes Damerau-Levenshtein distance on action ID sequences. Both metrics are reported to enable comparison.

5.3 Per-Task Breakdown

Table 4: Evaluation protocol reference: metric definitions, IoU thresholds, frame subsets, and tolerance settings for each task.

Task	Protocol source	Key parameter	Notes
ASD mAP	Schoonbeek 2024 [7] Table 3	mAP (b-boxed)	Evaluated on <i>annotated frames only</i> , not full videos. “mAP (entire videos) = 0.641” is a different protocol. Our implementation uses COCO 101-point all-point interpolation (mAP@[0.5:0.95]) as the primary metric.
ASD mAP@0.5	—	IoU= 0.5	Convenience metric on same frame set. Also report mAP@[0.5:0.95] (COCO standard).
Activity Top-1/5	Schoonbeek 2024 [7] §5.1	clip-level, 16 uniform frames	Single prediction per clip (majority vote or CLS read-out). POPW uses RGB modality only; MViTv2 multi-modal uses RGB+VL+stereo. See Section 5.2 for modality discussion.
PSR F1	PSRT paper [7]	± 3 -frame tolerance, full test set	POPW uses $\pm 3/\pm 5$ -frame tolerance bi-directional greedy matching per PSRT paper definition. Best PSRT baseline (B2) achieves F1=0.731 on full IndustReal test set; STORM-PSR achieves F1=0.506. The PSRT supplementary reports B3 F1=0.883 on error-free recordings. POPW’s PSR F1 is directly comparable to the PSRT paper protocol.
PSR Edit Score	PSRT paper [7]	Damerau-Levenshtein distance on state-change sequences, GT-normalized	Per-component OSA Damerau-Levenshtein on binary state strings, normalized by GT length (not max length). Matches STORM-PSR convention $(1 - \text{dist}/\text{len}(\text{GT}))$. The Edit Score is distinct from POS (run-based); baseline values differ across metrics.
PSR POS	PSRT paper [7]	Runs-based adjacent pair ordering	PSRT paper defines POS as fraction of correctly ordered adjacent pairs. Best PSRT baseline (B2) achieves POS=0.816. Note: STORM-PSR’s

Table 5: Activity recognition: per-protocol breakdown.

Dataset	Method	Top-1	Top-5	mcAP
IKEA-ASM (front, RGB)	P3D [1]	60.40	—	—
IKEA-ASM (front, RGB+pose)	I3D [1]	64.15	—	—
IKEA-ASM (all views)	I3D [1]	47.00	—	—
IKEA-ASM (all views)	PC3D [3]	80.20	—	—
IKEA-ASM (online, csv)	MiniROAD [4]	—	—	80.84
IKEA-ASM	POPW (ours)	—	—	—
<i>TVSeries cross-view OAD (PTMA benchmark, not IKEA ASM) [4]</i>				
TVSeries (cross-subject)	PTMA	—	—	84.47
TVSeries (cross-subj.+view)	PTMA	—	—	86.99
IndustReal	MViTv2 [7]	66.45	88.43	—
IndustReal	POPW (ours)	—	—	—

5.4 Efficiency Comparison

Table 6 compares the number of parameters, FLOPs, and throughput of POPW against published baselines on representative hardware.

Table 6: Efficiency comparison: parameters, FLOPs, and frames-per-second (FPS) on representative GPUs.

Dataset	Method	Params (M)	GFLOPs	FPS	Hardware
<i>IKEA-ASM</i> — 640×480					
	MiniROAD [4]	10.5	1.08	325	—
	PTMA [4]	12.9	1.96	291	—
	ActionFormer (RGB) [6]	27.7	83.28	21	GTX 1080
	Gated SRM [5]	33.5	121.09	16	GTX 1080
	POPW (ours, batched)	—	—	—	RTX 3060 [†]
	POPW (ours, streaming)	—	—	—	RTX 3060 [†]
<i>IndustReal</i> — 1280×720					
	YOLOv8m [7]	—	—	—	—
	MViTv2 [7]	—	—	—	—
	STORM-PSR [8]	—	—	—	—
	POPW (ours, batched)	—	—	—	RTX 3060 [†]
	POPW (ours, streaming)	—	—	—	RTX 3060 [†]

[†] POPW measured on RTX 3060 (12GB) with batch size 1 for streaming and batch size 8 for batched inference.

5.5 Ablation Studies

5.5.1 Ablation: Backbone Choice

[Table: ConvNeXt-Tiny vs. ResNet-50 vs. MobileNetV3 on detection AP, pose PCK, activity Top-1, PSR F1 — to be completed]

5.5.2 Ablation: Head Contributions

[Table: Detection-only, Pose-only, Activity-only, PSR-only, All heads (POPW) on each metric — to be completed]

5.5.3 Ablation: FiLM Conditioning

[Table: No FiLM, PoseFiLM only, HeadPoseFiLM only, Both FiLM stages on activity Top-1 — to be completed]

5.5.4 Ablation: Multi-Task Weighting

[Table: Equal weights, Human-tuned, Kendall uncertainty weighting on detection AP and activity Top-1 — to be completed]

5.6 Qualitative Results

[Figure: Qualitative results on IKEA ASM — showing detection bounding boxes, body pose overlays, activity predictions, and phase labels — to be generated]

[Figure: Qualitative results on IndustReal — showing ASD detections, head pose axes, activity predictions, and PSR component predictions — to be generated]

[Figure: Failure cases — showing occluded assemblies, ambiguous activity classes, and PSR errors — to be generated]

6 Discussion and Analysis

6.1 Trade-offs: Accuracy vs. Efficiency

The design of POPW reflects a fundamental trade-off in multi-task learning: sharing features across tasks reduces computation but may degrade per-task accuracy compared to dedicated models. A single YOLOv8m + MViTv2 + STORM-PSR pipeline requires three separate backbones, each extracting features independently. POPW’s shared backbone eliminates this redundancy, reducing total parameters and FLOPs.

However, the shared representation must capture task-agnostic features useful for all heads, which may require larger backbone capacity than any single-task model. The efficiency gains from sharing must be weighed against potential accuracy losses from representation constraints.

6.2 Which Tasks Benefit Most from Shared Representation?

The FiLM conditioning mechanism enables cross-task information flow that would not be possible with purely shared features. Detection context (f_{det}) provides object state information to activity recognition, helping distinguish similar poses in different assembly contexts. Similarly, head pose conditioning captures gaze direction, which is critical for understanding where the worker’s attention is focused.

The PSR head maintains an independent input path (multi-scale FPN features), avoiding interference from other tasks. This architectural choice reflects the intuition that procedure step recognition requires different visual cues than detection or activity recognition.

[Quantitative analysis of cross-task benefits to be completed]

6.3 Failure Cases

Multi-task models may fail in ways that single-task models do not. Gradient interference can cause tasks to degrade each other’s performance. If one task’s loss dominates early in training, it may push the backbone to representations that are suboptimal for other tasks.

[Analysis of failure modes: (1) heavily occluded assemblies where ASD confidence drops, (2) rare action classes with <10 training examples, (3) cross-view generalization on IKEA ASM when only trained on front view]

7 Conclusion

We presented POPW, a unified multi-task architecture for egocentric assembly understanding that performs object/state detection, body pose estimation, head pose tracking, activity recognition, and procedure step recognition in a single forward pass.

1. **Unified architecture:** A shared ConvNeXt-Tiny + FPN backbone with five task-specific heads achieves competitive accuracy against dedicated single-task baselines while eliminating redundant feature extraction.
2. **FiLM conditioning:** Two-stage FiLM modulation (PoseFiLM and HeadPoseFiLM) effectively propagates pose and viewpoint information into activity recognition features without gradient interference.
3. **Multi-task loss balancing:** Kendall homoscedastic uncertainty weighting combined with staged training enables stable joint optimization across heterogeneous tasks.

7.1 Limitations

[Limitation 1: POPW requires higher resolution input (1280x720) than some single-task baselines, limiting deployment on memory-constrained edge devices.]

[Limitation 2: The head pose head is specific to IndustReal and cannot be directly transferred to other egocentric datasets without ground-truth head pose annotations.]

[Limitation 3: PSR performance degrades on videos longer than $MAX_LEN=32$ frames due to fixed-length feature bank; long-horizon temporal reasoning is limited.]

[Limitation 4: Staged training adds complexity; end-to-end training from scratch results in gradient interference and XXX% lower final performance.]

[Limitation 5: YOLOv8m, MViTv2, and STORM-PSR are from different institutions and trained independently; POPW’s shared backbone may not fully capture the domain-specific inductive biases of each task.]

[Limitation 6: Code for PTMA and MiniROAD is not publicly available, making direct reproducibility comparisons incomplete.]

7.2 Future Work

[Future work 1: Investigate adaptive feature bank length for long-duration assembly videos with variable step counts.]

[Future work 2: VideoMAE V2 fusion is implemented (see §2.2.4); future work includes TimeSformer or VideoMAE in early layers for even stronger temporal modeling.]

[Future work 3: Explore knowledge distillation from single-task teacher models into the multi-task POPW to recover accuracy lost to task interference.]

[Future work 4: Extend to cross-domain generalization: train on IndustReal synthetic data and fine-tune on real-world factory assembly.]

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